# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 2

**Full Syllabus (Chapters 1–13)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** The magnetic field produced by a straight wire at a distance 4 cm is B. At 2 cm from the wire (same current), the field is 1
**(a)** smaller than B **(b)** equal to B **(c)** zero **(d)** larger than B

**2.** Which of the following is the correct food chain? 1
**(a)** Plant, caterpillar, bird, hawk **(b)** Hawk, bird, caterpillar, plant
**(c)** Bird, plant, hawk **(d)** Plant, hawk, bird

**3.** In $\mathrm{ZnO + C \longrightarrow Zn + CO}$, the reducing agent is: 1
**(a)** ZnO **(b)** Zn **(c)** CO **(d)** C

**4.** $\mathrm{2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2}$ is a: 1
**(a)** combination **(b)** displacement
**(c)** thermal decomposition **(d)** neutralisation

**5.** Electronic configuration of an element X is 2, 8, 2. It forms an ion of charge: 1
**(a)** -2 **(b)** +1 **(c)** +2 **(d)** 0

**6.** $\mathrm{CH_3CH_2OH}$ with alkaline $\mathrm{KMnO_4}$ and heat gives: 1
**(a)** $\mathrm{CH_3COOC_2H_5}$ **(b)** $\mathrm{CH_3COOH}$ **(c)** $\mathrm{CH_2{=}CH_2}$ **(d)** $\mathrm{CH_3CH_3}$

**7.** Anaerobic respiration releases energy in the: 1
**(a)** cytoplasm **(b)** chloroplast **(c)** mitochondria **(d)** ribosome

**8.** If the tip of a shoot is removed, phototropic bending: 1
**(a)** increases **(b)** does not occur **(c)** occurs more **(d)** reverses

**9.** A flower has only stamens. It is 1
**(a)** sterile **(b)** female **(c)** bisexual **(d)** unisexual (male)

**10.** A couple already has two daughters. The probability that their third child is a boy is 1
**(a)** 1/2 **(b)** 1/4 **(c)** 1 **(d)** 0

**11.** Speed of light in a medium of n = 2 is 1
**(a)** $1.5 \times 10^8$ m/s **(b)** $3 \times 10^8$ m/s **(c)** $6 \times 10^8$ m/s **(d)** $2 \times 10^8$ m/s

**12.** The deviation of violet light in a prism is greater than that of red because 1
**(a)** violet has lower frequency
**(b)** the refractive index for violet is higher
**(c)** red is absorbed
**(d)** violet travels faster in glass

**13.** A 10 ohm and a 40 ohm resistor are in parallel. The equivalent resistance is 1
**(a)** 25 ohm **(b)** 8 ohm **(c)** 50 ohm **(d)** 5 ohm

**14.** An alpha particle moving north is deflected towards the east in a magnetic field. The force is 1
**(a)** perpendicular to the velocity **(b)** against the velocity
**(c)** along the velocity **(d)** zero

**15.** Which of the following would show the greatest biomagnification? 1
**(a)** A hawk that eats snakes **(b)** Grass
**(c)** A grasshopper **(d)** A snail

**16.** Hydrogen and oxygen are obtained in the ratio 2 : 1 by volume because: 1
**(a)** oxygen is more soluble
**(b)** hydrogen is lighter
**(c)** oxygen is heavier
**(d)** water contains 2 hydrogen atoms for each oxygen atom

**17.** Which gas is evolved when $\mathrm{NaOH}$ reacts with zinc? 1
**(a)** chlorine **(b)** carbon dioxide **(c)** hydrogen **(d)** oxygen

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**18.** Zinc coating protects iron even if scratched because zinc: 1

**(a)** is less reactive than iron **(b)** is heavy
**(c)** is more reactive than iron **(d)** is non-metallic

**19.** Carbon cannot form $C_4^+$ because: 1

**(a)** it has 2 valence electrons
**(b)** it has 8 valence electrons
**(c)** high energy is needed to remove 4 electrons
**(d)** it is a gas

**20.** Normal blood pressure (systolic/diastolic) in human beings is: 1

**(a)** 100/100 mm Hg **(b)** 120/80 mm Hg **(c)** 80/120 mm Hg **(d)** 140/90 mm Hg

**21.** A child of age 12 is very short but healthy. The hormone lacking: 1

**(a)** insulin **(b)** thyroxine only **(c)** growth hormone **(d)** adrenaline

**22.** The zygote in humans has 1

**(a)** 23 chromosomes **(b)** 12 chromosomes **(c)** 92 chromosomes **(d)** 46 chromosomes

**23.** A bacterium reproducing asexually will show variation because of 1

**(a)** meiosis in parents **(b)** gamete fusion
**(c)** crossing of two parents **(d)** minor DNA copying errors

**24.** An object 3 cm high is at 20 cm from a convex lens of f = 10 cm. The image height is 1

**(a)** 6 cm **(b)** 3 cm (inverted) **(c)** 3 cm (erect) **(d)** 1.5 cm

**25.** The Sun at noon appears white because 1

**(a)** light travels a short path and little blue scatters
**(b)** red gets scattered
**(c)** violet is absorbed
**(d)** no scattering occurs

**26.** A 100 W bulb has a resistance at 220 V of 1

**(a)** 22 ohm **(b)** 100 ohm **(c)** 2.2 ohm **(d)** 484 ohm

**27.** The induced current in a coil is zero when 1

**(a)** the magnet is moved
**(b)** the magnet is at rest relative to the coil
**(c)** the coil is moved
**(d)** the field changes

**28.** A sewage treatment plant converts waste into 1

**(a)** plastic **(b)** fuel only
**(c)** manure and clean water **(d)** glass

**29.** The coefficient of $O_2$ in the balanced equation $KClO_3 \longrightarrow KCl + O_2$ is: 1

**(a)** 4 **(b)** 2 **(c)** 1 **(d)** 3

**30.** Which of the following pairs of solutions will produce a neutral salt solution? 1

**(a)** $H_2CO_3$ and NaOH **(b)** HCl and NaOH **(c)** $CH_3COOH$ and NaOH **(d)** HCl and $NH_4OH$

**31.** Which of the following reactions will take place? 1

**(a)** $Cu + ZnSO_4 \longrightarrow CuSO_4 + Zn$ **(b)** $Au + FeSO_4 \longrightarrow Au_2(SO_4)_3 + Fe$ **(c)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(d)** $Ag + CuSO_4 \longrightarrow Ag_2SO_4 + Cu$

**32.** The number of shared electron pairs in ethene ($C_2H_4$) is: 1

**(a)** 6 **(b)** 8 **(c)** 5 **(d)** 4

**33.** Stomata are open during the day because: 1

**(a)** guard cells lose water **(b)** leaves fall
**(c)** guard cells take up water and swell **(d)** light is absent

**34.** A person touches a hot plate and quickly pulls the hand. The effector is: 1

**(a)** skin **(b)** muscle **(c)** brain **(d)** spinal cord

**35.** Which statement about variation is correct? 1

**(a)** DNA copying is always perfect
**(b)** Variation harms every species
**(c)** Inaccurate DNA copying produces variation
**(d)** Only sexual reproduction ever has variation

**36.** Out of 160 F2 plants in a monohybrid cross, the expected number of dwarf plants is 1

**(a)** 80 **(b)** 120 **(c)** 20 **(d)** 40

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**37.** An object is placed 18 cm from a concave mirror of f = 12 cm. The image distance is [1]

**(a)** 36 cm **(b)** 24 cm **(c)** 18 cm **(d)** 12 cm

**38.** A lens of power -2.5 D has focal length [1]

**(a)** -2.5 cm **(b)** +40 cm **(c)** -25 cm **(d)** -40 cm

**39.** The work done in moving 4 C of charge across a potential difference of 9 V is [1]

**(a)** 13 J **(b)** 36 J **(c)** 2.25 J **(d)** 5 J

**40.** Looking at a circular coil, the current is anticlockwise. The face seen behaves as [1]

**(a)** a north pole **(b)** neutral **(c)** a south pole **(d)** both poles

**41.** If the producers of a food chain have 50,000 J of energy, the tertiary consumers receive [1]

**(a)** 500 J **(b)** 50 J **(c)** 5000 J **(d)** 5 J

**42.** $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ is a: [1]

**(a)** displacement **(b)** precipitation

**(c)** photolysis **(d)** thermal decomposition

**43.** On passing excess $CO_2$ in lime water, the milkiness: [1]

**(a)** turns pink **(b)** disappears **(c)** remains same **(d)** increases

**44.** An element with configuration 2, 8, 7 forms an ion: [1]

**(a)** $X^{+}$ **(b)** $X^{2+}$ **(c)** $X^{2-}$ **(d)** $X^{-}$

**45.** Which of the following will decolourise bromine water? [1]

**(a)** $CH_4$ **(b)** $C_2H_4$ **(c)** $C_2H_6$ **(d)** $C_3H_8$

**46.** Compared with anaerobic respiration, aerobic respiration releases: [1]

**(a)** no energy **(b)** the same energy

**(c)** much less energy per molecule of glucose **(d)** much more energy per molecule of glucose

**47.** Which hormone is responsible for the wilting of leaves? [1]

**(a)** auxin **(b)** cytokinin **(c)** gibberellin **(d)** abscisic acid

**48.** An ovary has 20 ovules and all of them are fertilised. The fruit will have [1]

**(a)** 20 seeds **(b)** 0 seeds **(c)** 1 seed **(d)** 2 seeds

**49.** Which of the following statements about sex determination is correct? [1]

**(a)** Egg decides the sex **(b)** Mother's age decides

**(c)** Both eggs and sperm always give girls **(d)** Sperm decides the sex

**50.** Light enters from air into a medium at an angle of 45 degrees and refracts at 30 degrees. The refractive index is [1]

**(a)** $\sqrt{3}$ **(b)** $1/\sqrt{2}$ **(c)** 2 **(d)** $\sqrt{2}$

**END OF QUESTION PAPER** **Total: 50 Marks**

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